Ciphix RPA Academy

Robotic Process Automation – RPA Applicants Course

RPA Applicants Course: Introduction to UiPath

C1PH1X

Let’s start your RPA journey together!

Are you ready?

C1PH1X

Introduction

Introduction to UiPath – What to Expect?

Welcome to the lesson on Introduction to UiPath. In this part of the RPA Applicants course you will be introduced to the basic concepts of UiPath and start building your first robots! Every chapter starts with an introduction video, including the theory of the topic followed by an example in UiPath. After watching an introduction video, there are a couple of slides with a more in-depth explanation of the theory. After you learned all the important theory you will end the chapter by doing a practical exercise. In this way you will learn how to implement the knowledge you just learned in UiPath. To help you understand everything perfectly a solution video is provided with every exercise. Don't forget the tips & tricks you just learned in the previous part of this course while running into challenges. At the end of these lessons, you are ready to tackle your RPA Case!

Enjoy and Good Luck!

"Innovate Fearlessly"

Ciphix RPA Academy – Introduction to UiPath

01 UiPath Studio

02 Data Types & Variables

03 Activities

04 Conditional Activities

05 Data Tables

06 Sequences & Flowcharts

07 Selectors

08 RPA Case

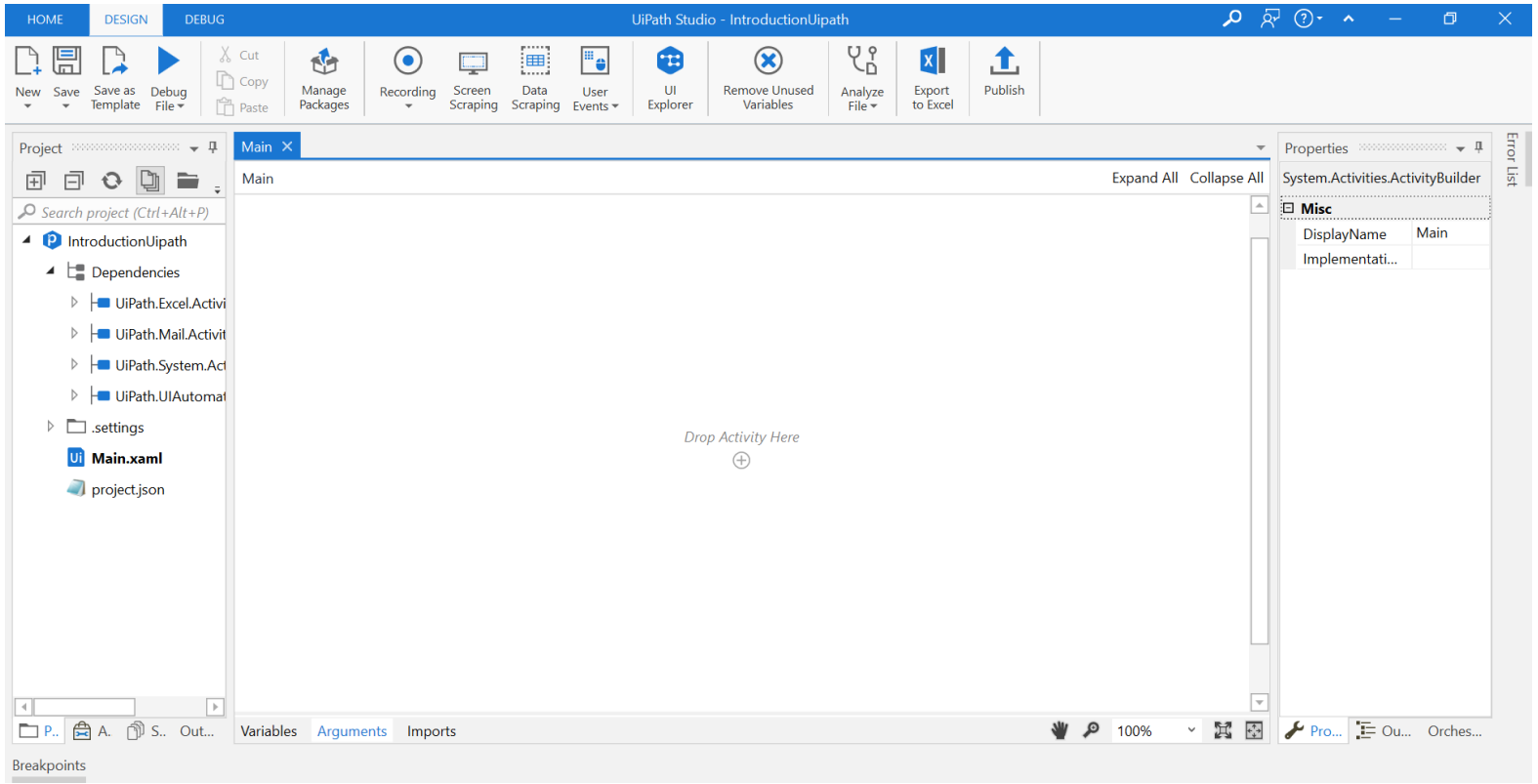
Lesson 3: UiPath Studio

Guide Through UiPath Studio

Lesson 3 - UiPath Studio

UiPath Studio – Introduction Video

In this chapter you will be introduced to the UiPath Studio where we design our robots. Start here with the Introduction video which explains what UiPath Studio is, how you can create a new project within UiPath Studio and walks you through the interface of the Studio.



[UiPath Studio Introduction Video]

Lesson 3 - UiPath Studio

UiPath Studio - Recap

This chapter introduced you to the UiPath Studio: the interface where we design, build, and test our robots. By now you're getting a bit more familiar with the UiPath interface, but it is time to start getting to know the tools in UiPath more in depth. Let's start with the first step. You want the robot to interpret certain information and does something with that information. How can you make sure your robot understands this data correctly? Let's see how this works in the next chapter about data and variables!

Definition of UiPath Studio

How to work with UiPath Studio

Lesson 4: Data Types & Variables

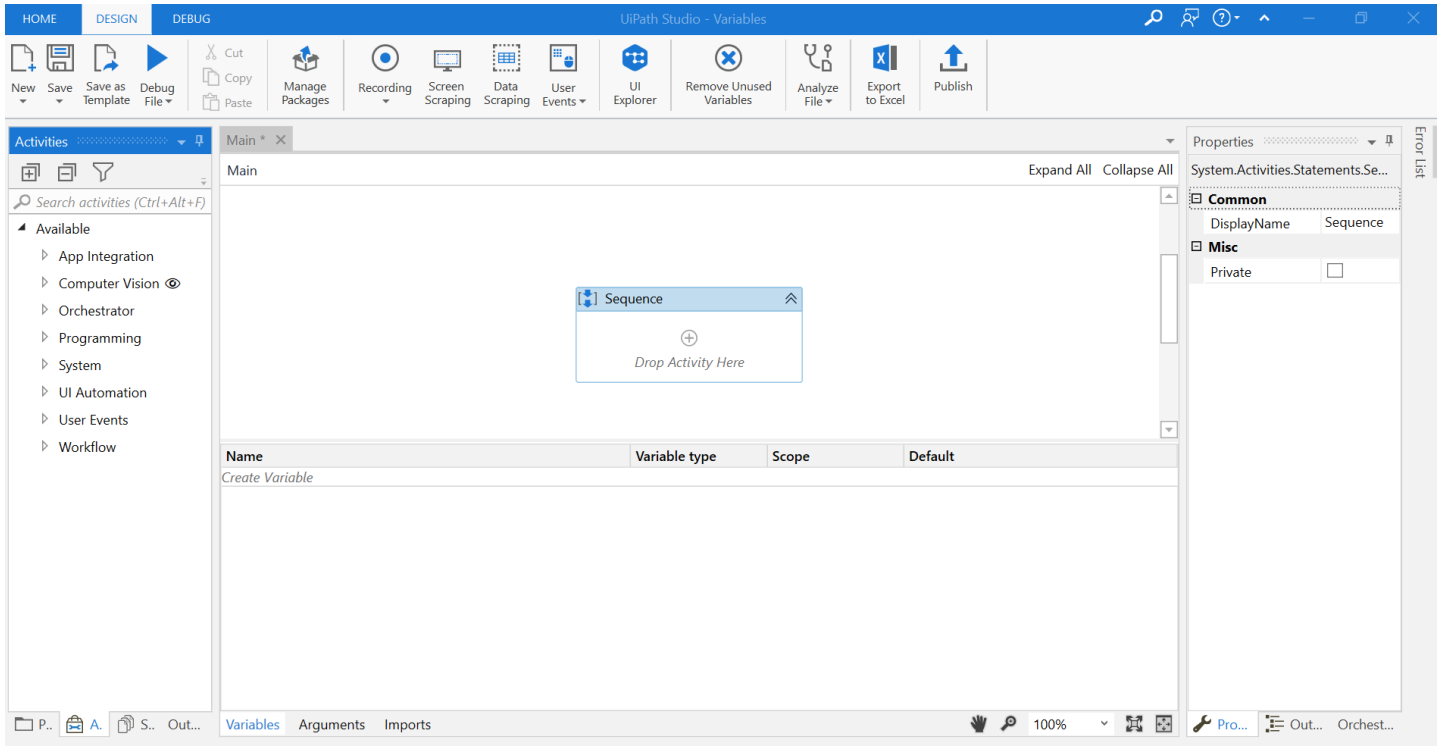
Introduction to Data, Data Types & Variables

Lesson 4 - Data Types & Variables

Data Types & Variables – Introduction Video

In this chapter of the course you will be introduced to data types and variables. While building robots within UiPath Studio you will work a lot with these concepts, therefore it is important that you start understanding all data types and variables before starting to build your first robot. Start with the Introduction video to understand what these concepts mean and how you can use them in automating your processes. There will be an exercise at the end of this chapter so pay attention!

Good Luck!



[Introduction Video Data Types & Variables]

Lesson 4 - Data Types & Variables

Data types – Introducing Data

What? Data is all information that we use in our process. For example, an invoice number or an email address.

Why? Robots don't know what they must insert when we want information to be filled in in certain applications/actions. As a developer, you need to feed the robot with **Data** to let it perform actions.

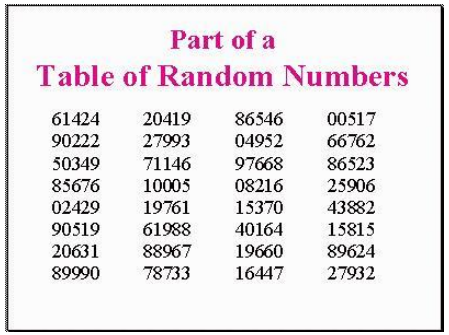
How? Data comes in different shapes and sizes, the datatype. A **data type** determines what type of value a data object can have and in which operations it can be used. Data can only have one data type.

Lesson 4 - Data Types & Variables

Data Types – What is a data type?

Data comes in different shapes and sizes. A **data type** determines what type of value a data object can have and what operations it can perform. Common datatypes include:

- Text
- Numbers
- Images
- True/False
- Data Tables



Part of a
Table of Random Numbers

61424	20419	86546	00517
90222	27993	04952	66762
50349	71146	97668	86523
85676	10005	08216	25906
02429	19761	15370	43882
90519	61988	40164	15815
20631	88967	19660	89624
89990	78733	16447	27932

It was the best of times, it was the worst of times, it was the age of wisdom, it was the age of foolishness...

Note: It is important to understand that your computer **treats every type of data differently**. For example: your computer can work with numbers like a calculator: *1 + 1 = 2*. But if your numbers are actually stored as text, your computer will treat your numbers as text: *"1" + "1" = "11"*.

Lesson 4 - Data Types & Variables

Data Types - Integer and Double

- An **integer** is a numeric value without a decimal.
- Integers are whole numbers and can be positive or negative.
- A number with a decimal is referred to as a **Double**.
- Examples:

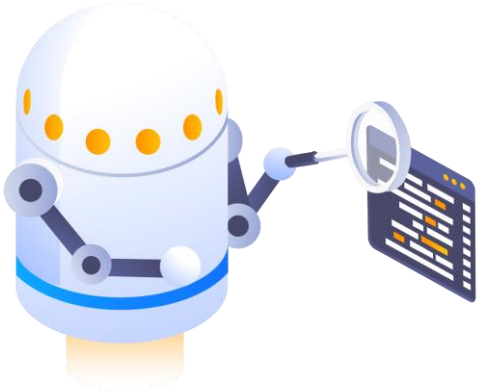
Integer:

- 24
- 1,450
- -5

Double:

- 4.5
- 1.574635
- -3.87

Note: It is important that integers and double data types can only consist of numbers. So when you have a date 01-02-2020, this can not be of type integer. Instead, this is of type string. When the data is 01022020 it can be of type integer.



Lesson 4 - Data Types & Variables

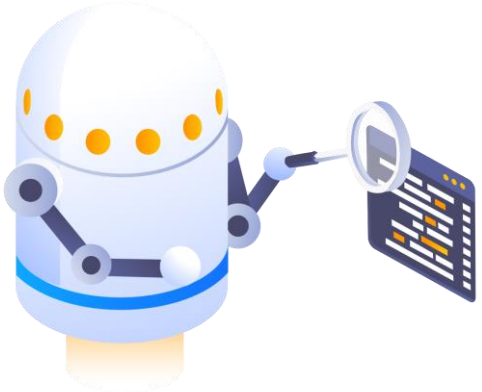
Data Types - String

One of the mostly used data types is a **string.** A string consists of one or more characters, which can include letters, numbers, and other types of characters.

- A string represents alphanumeric data. This means that a string can contain many different characters, but they are all considered as if they were text, even if the characters are numbers.
- A string can also contain spaces.
- Strings are distinguished by quotation marks **""**, these ensure that the text in between them is recognized as a string and not as separate values or other data types.
- **Examples:**

- “Hello World” = Hello World
- “73rtgiewpdjwhe927rgd&#\\” = 73rtgiewpdjwhe927rgd&#\\

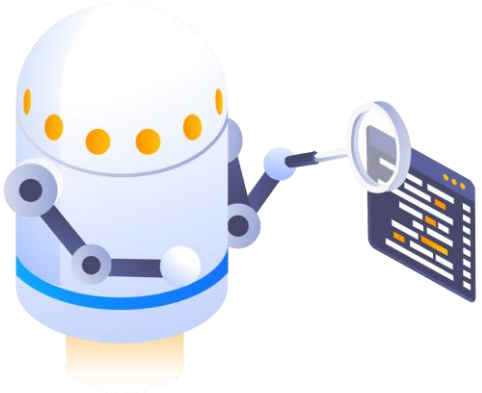
- The use of quotes also makes it possible to use numeric characters as part of a string.
- For example, the following code is used to store a street address: “Roboroad 123”. Without the quotes, the numeric characters 123 would be interpreted as a number, but with the quotes, it is recognized as being part of a string that represents a street address.



Lesson 4 - Data Types & Variables

Data Type - Boolean

- The **Boolean** data type can only represent two values: true or false.
- These variables enable you to make decisions, and thus have a better control over your flow.
- Boolean variables are often used in conditional statements
- **Example:**
- If an animal has hair it is specified as hairy. A Boolean variable would store True or False whether an animal is hairy or not. So when an animal is a dog this Boolean variable is true. When this animal is a chicken this Boolean variable is false.



Lesson 4 - Data Types & Variables

Data Type - Array

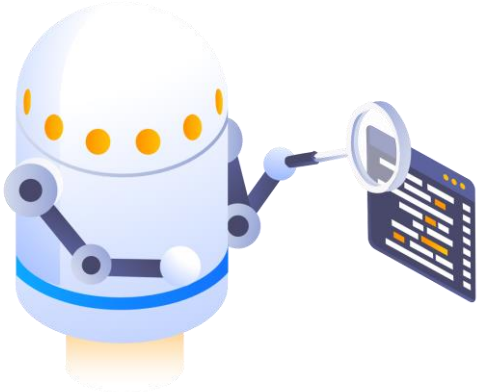
The array variable is a type of variable which enables you to store multiple values of the same data type.

UiPath Studio supports as many types of arrays as it does types of variables. This means that you can create an array of integers, one of strings, one of Boolean values and so on.

Arrays can only contain data with **the same** data type

Examples:

- Array of strings containing multiple animal species: {“Horse”, “Dog”, “Panda”, “Penguin”}
- Array of integers containing multiple postcodes: {3033, 4566, 7688, 1516}

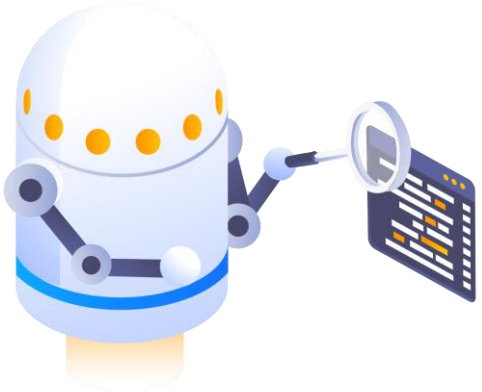


Lesson 4 - Data Types & Variables

Data Type - Generic Value

The GenericValue variable is a type of variable that can store **any kind of data**, including text, numbers, dates, and arrays, and is particular to UiPath Studio

GenericValue variables are automatically converted to other types, in order to perform certain actions. However, it is important to keep in mind that their conversion is unpredictable and may not always be the correct one for your project. Therefore, it is **not recommended** to use this data type and specify the right data type for each value



Lesson 4 - Data Types & Variables

Data Types - Summary of Important Data Types

Type	Description	Example
Generic Value	Undetermined data type. UiPath will guess the type when it is used. This type is not recommended.	Any
String	Text values that must always be between double quotes.	“A piece of text”
Integer (Int32)	Number value. 32 denotes the maximum bit size of the variable.	12345
Boolean	A variable that can only hold True or False and can be used to make decisions.	pageExists = False
Array<T>	A list variable that stores the type of data that is defined between the brackets <T>.	{‘apple’,’banana’,’orange’}
Data Table	A table variable with rows and columns. Columns can also be named. (Will be discussed in chapter 5)	An extracted Excel file

Lesson 4 - Data Types & Variables

Variables – Introducing Variables

What?

A **variable** can be described as a box that can be filled with information. This information can be placed inside the box by hand or by using UiPath Activities. Also, UiPath has numerous Activities to change, replace, move or delete information inside the box (variable).

Why?

Variables provide a way of labeling data with a descriptive name, so our programs can be understood more clearly by the reader and ourselves. This way you can find the variable in your process and retrieve and use the information from inside the variable later in the process. Furthermore, variables provide us with the possibility of dynamically entering our data. This makes the maintainability of our robots much higher

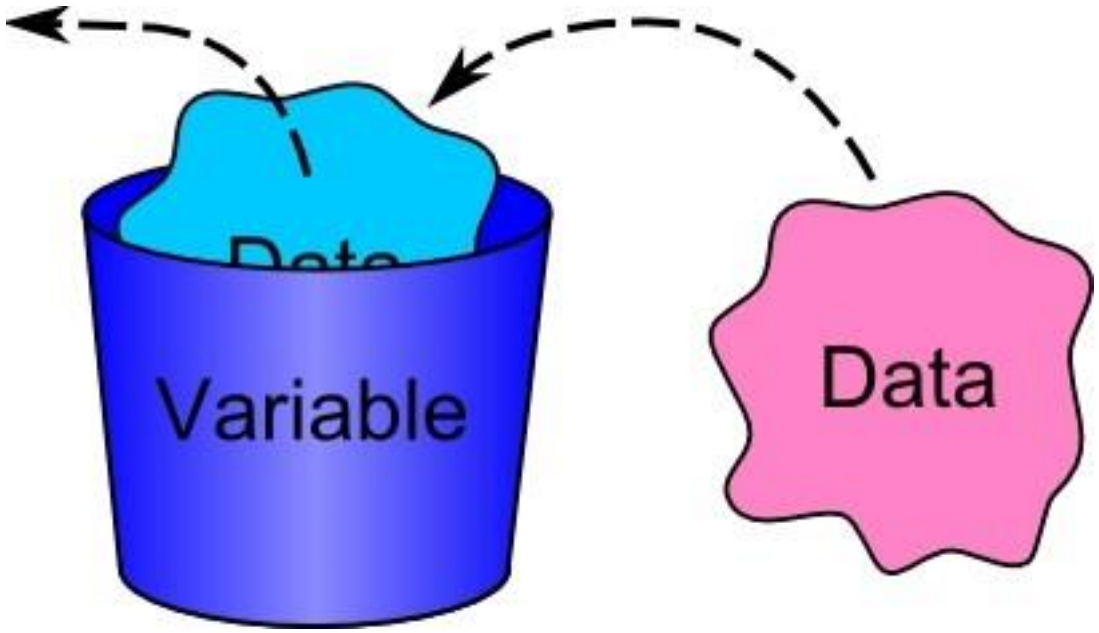
How?

Variables are used to store data that can be referenced and manipulated within a workflow. We can create variables in the studio in three ways – the variable panel, the designer panel, and the properties panel

Lesson 4 - Data Types & Variables

Variables – Examples of Variables

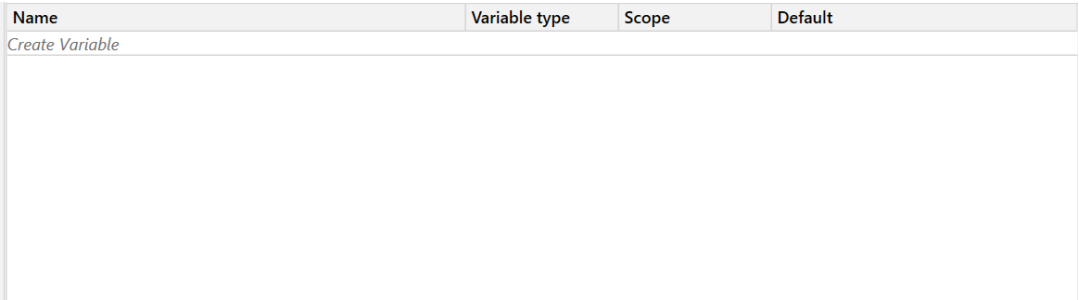
- **dateOfBirth** can be a variable of type **string** that holds the value of a date of birth e.g. “24-03-1995:
- **websiteURL** can be a variable of type **string** that holds the value of a URL e.g. “https://www.google.com”
- **phoneNumbers** can be a variable of **array{string}** that holds the value of three phonenumber e.g. {“06-12345678”, “06-45633893”, “06-98372056”}
- **monthlyIncome** can be a variable of type **integer** that holds the value of a monthly income e.g. 1500



Lesson 4 - Data Types & Variables

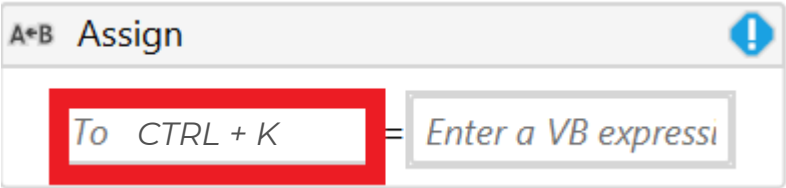
Variables - Creating Variables

There are three ways to create variables in UiPath:

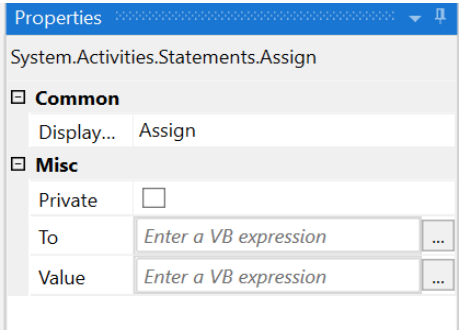


1. From the **Variable pane** - Open the Variables pane, select the 'Create new Variable' option. Select the right datatype for this variable, with clicking on the variable type. If you do not see your type in the row, select browse types and click on the right type. Chose the scope in which you want to use the variable and set a Default value if necessary.

2. From the **Designer pane** - Drag an activity with a variable field visible (i.e. 'Assign') and press Ctrl+K in the field where you need the variable. Name it and then check its properties (Type and Scope) in the Variables pane



3. From the **Properties pane** - In the Properties pane of the activity, place the cursor in the field in which the variable is needed (i.e. Output) and press Ctrl+K. Name it and then check its properties in the Variables pane.



Lesson 4 - Data Types & Variables

Variables - Naming Variables

A variable is made up of two parts: a descriptive and unique **name** and a **value**.

A good variable name must adhere to the following guidelines:

- **camelCase** : the first letter of every word is in capital, indicating that it is a new word, except for the first letter of the variable name.
- **Data types** : When using variables in UiPath the type of variable needs to be mentioned at the beginning of the variable in lower case with a shortcode (max length: 4 characters). Shortcodes can be: Datatable(dt), Integer(int), Array (arr), String(str)

The format for a variable name is: < Variable Short Code >< Variable Description >

Important with naming Variables:

- **string / integer / array / boolean / etcetera**: these are datatypes, not variable names. By using these variables, the name is by definition not descriptive and not unique.
- **data**: by using the variable name data, it is unclear what data is meant.

Good Names	Reason Good	Bad Names	Reason Bad
intIndexRowExchangeDT	The variable is very clear in what it is used for and makes good use of the camelCase-rule. The DT does not have to be at the start here, because it does not concern a DataTable.	index	The variable name does not indicate what it is used for, other than that it is an index.
dtExchange	The variable name immediately makes it clear that it is a datatable and what that datatable is used for.	dt1	It is not clear by the name of this variable what the datatable is used for. Furthermore, the 1 in the name does not make it unique.
arrArgumentProperties	The variable name is clear in what it is used for, makes good use of camelCasing, and declares the special type.	RecordsArray	The variable name does not use correct camelCasing, is not really descriptive in what records it uses, and has to start with array.
intArticleNumber	The variable name is clear in what it is used for and makes good use of camelCasing.	AN	The variable name does not make it clear what the variable is used for, this abbreviation could stand for a lot of things.
isAddressAvailable	The variable name is clear in what it is used for and makes good use of camelCasing	available	The variable name does not indicate what is available.

Table 1.1: Examples of good and bad names for variable

Lesson 4 - Data Types & Variables

Variables - Selecting a Data Type For Your Variable

You can manage the kind of data you want to store in your variable by assigning a **Variable Type** (see Variable Pane Figure). Now when you have created a variable of type String (text), UiPath will throw an error when you try to store an Integer (number) inside it. Therefore, it is important to select the right data type for your variable.

Name	Variable type	Scope	Default
Create Variable			

Name	Variable type	Scope	Default
variable1	String	Sequence	*Enter a VB expression*
variable2	Int32	Sequence	*Enter a VB expression*
variable3	Boolean	Sequence	*Enter a VB expression*
variable4	DataTable	Sequence	*Enter a VB expression*
Create Variable			

Lesson 4 - Data Types & Variables

Data Types & Variables - Exercise

In this assignment you're going to get familiar with UiPath data types and variables. You will create a set of variables that store different types of data.

Assignment

Create 3 different variables that contain the following information about your Pet:

- Species
- Name
- Monthly Expenses

- Make sure you use a different way of creating a variable every time!
- Make sure you assign the right **data type** to your variable

Time: 15 minutes

Solution Video

Lesson 4 - Data Types & Variables

Data Types & Variables - Recap

Good Job! You have learned all the important aspects of working with data and variables and finished your first UiPath assignment!

You now know what data is and that **data** can have different types. It is important to give your data a data type such that your robot knows how to correctly interpret this information. You also learned that **variables** work like containers, and that you can store data within a variable. In this way you can dynamically work with data without altering it within your workflow.

How can we create these variables? Easy! There is not one way to do that but three! Create a variable from the Variable Pane, the Designer Pane or from the Properties Pane of an activity and you are all set! So If you store this data in variables, you can use and change this data while you are working with activities.

But what do we mean with activities? Let's explore this topic in the next chapter!

- Definition of Data
- Specifying Data Types
- Definition of Variables
- Creating a Variable from the Variable Pane
- Creating a Variable from the Designer Pane
- Creating a Variable from the Properties Pane

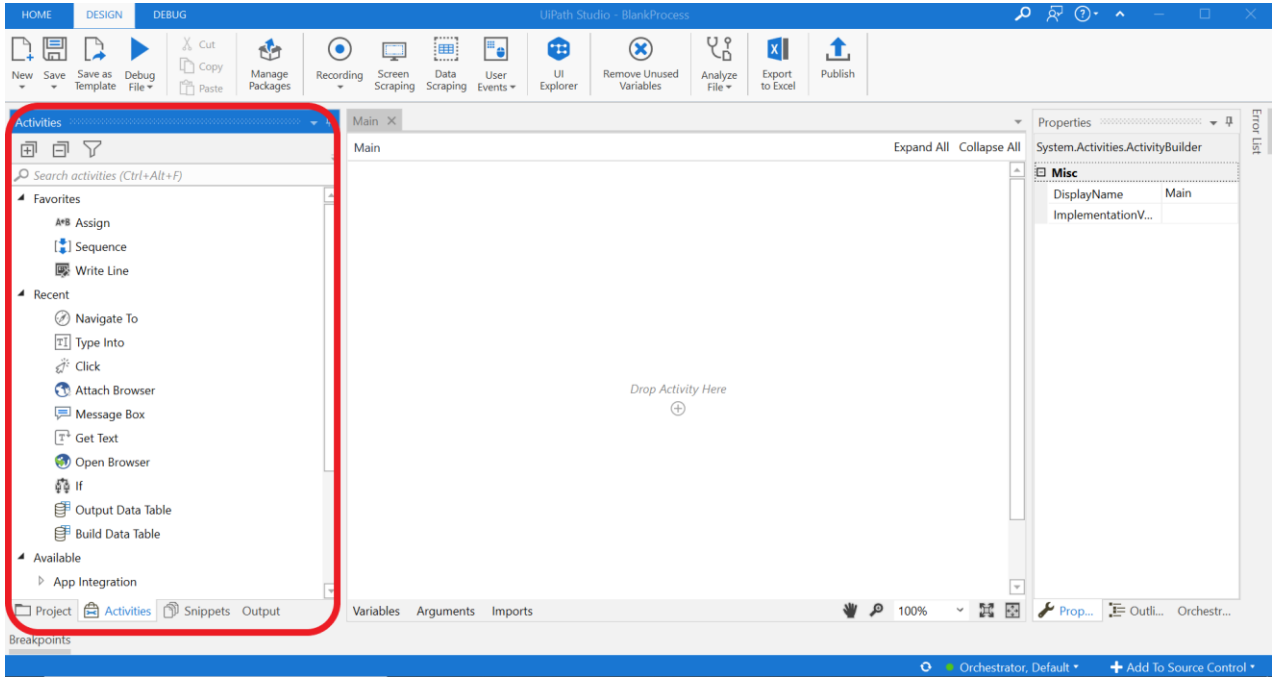
Lesson 5: Activities

Introduction to The UiPath Activities

Lesson 5 - Activities

Activities - Introduction Video

In this part of the course you will learn how to use Activities within UiPath studio. First, watch this video carefully to get familiar with the basic theory of activities. After this video, a couple of mostly used activities will be highlighted. Every activity also contains an introduction video that will explain what it does and for which actions you can use it. Make sure you watch them all, because at the end of this chapter there will be an exercise where you are going to use each activity in one process!



[Introduction video to Activities]

Lesson 5 - Activities

Activities – Introducing Activities

What? **Activities** are predefined instructions on what the robot needs to do.

Why? There are over a hundred different types of Activities in UiPath which all let your Robot perform different actions.

How? Select the Activity you need from the Activity Pane and drag it into your workflow. It is important to give your Activity a descriptive name including the action that the activity is performing. For example if you want your robot to click the save button in a word document you should name this Activity “Click the Save Button”. A descriptive name helps keeping your work understandable in the future.

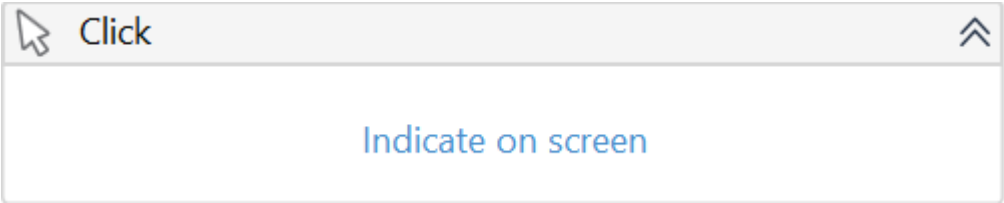
Lesson 5 - Activities

Click Activity

Arguably the most used Activity is the basic **Click** Activity. This activity is used to let the robot Mouse Click on anything you want, e.g Buttons, applications and text boxes.

When dragging this Activity in your Workflow it will display *Indicate on screen* in the centre of the block. By clicking on this blue text the Studio will minimize and the top-level window below the Studio will appear. Make sure this is the window you want to let your Robot interact with. When hovering your mouse over the window, blue boxes with a golden border will appear around all the different user interface elements (**UI elements**) that can be found in the window. Click on the element you want your Robot to click. The Studio will pop back up and the Click Activity will now display a screenshot of the element you have just clicked. Hit the Run button to see if your Robot does what you just programmed it to do.

Video - Click Activity

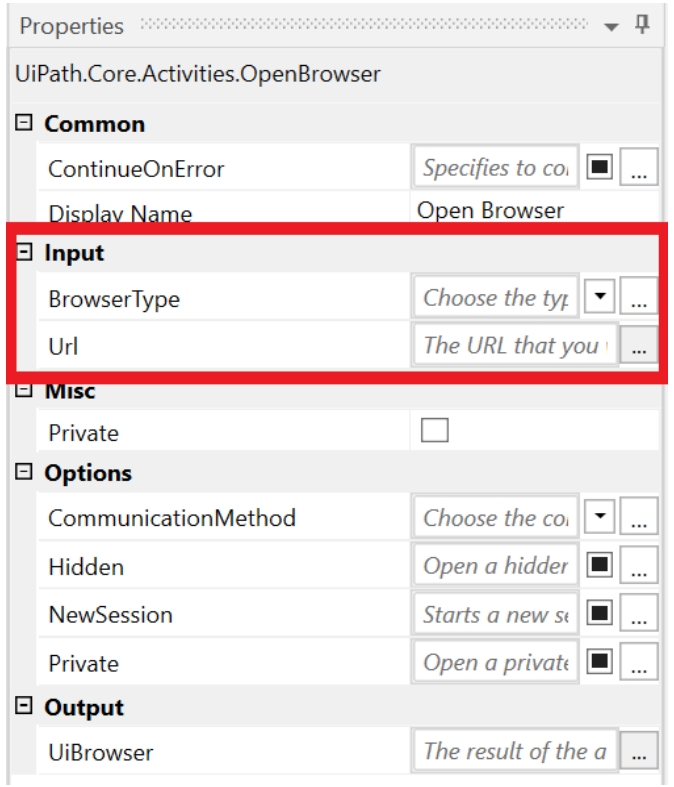


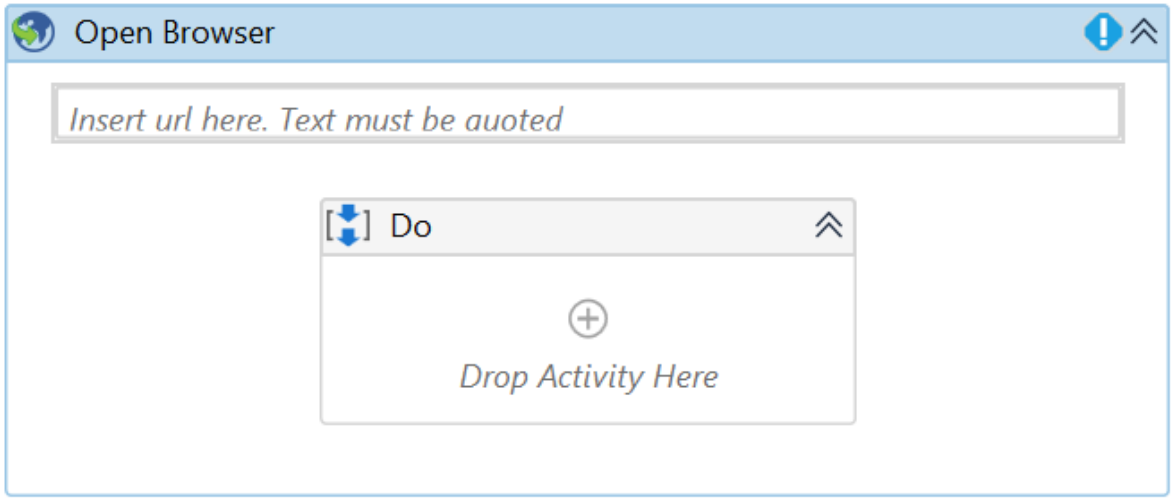
Lesson 5 - Activities

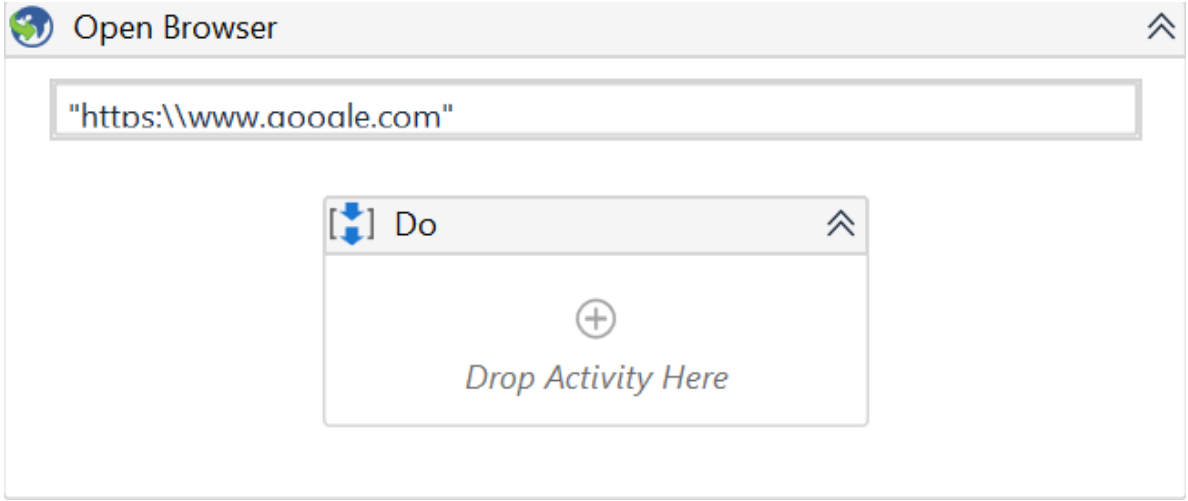
Open Browser Activity

Video - Open Browser Activity

The Open Browser Activity enables you to open a browser at a specified URL and execute multiple activities within it. The input property of this activity exists of Browsertype and URL. Here you select the browser that you are using, and can you insert the URL that you want to open in the chosen browser.







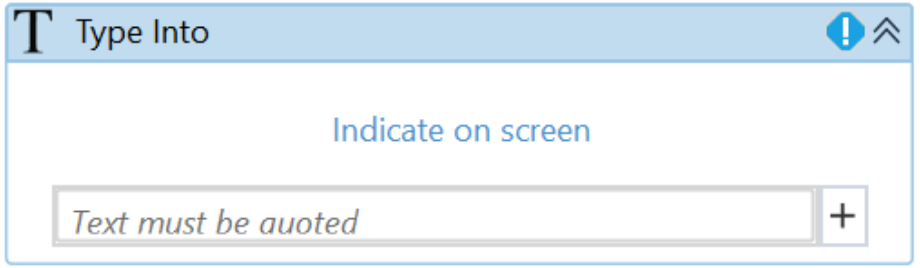
Lesson 5 - Activities

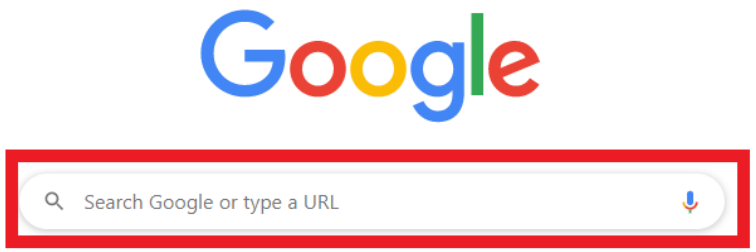
Type Into Activity

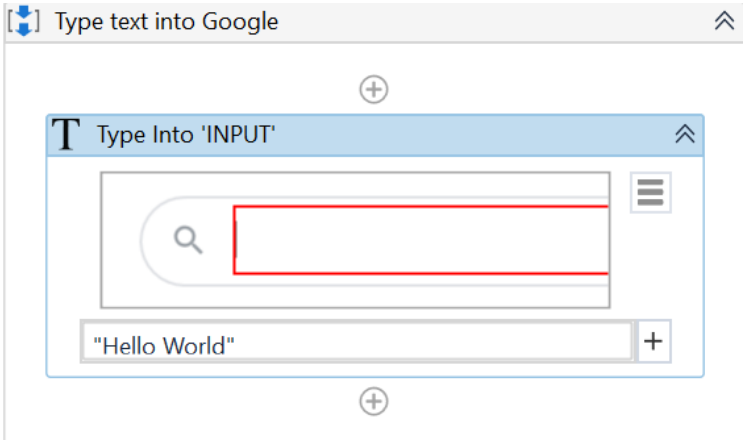
Video - Type Into Activity

With the type into activity you can **type text into UI Elements**. Examples are typing your login credentials inside Facebook, type an address in Google Maps or type a specific question in Google. When dragging this Activity in your Workflow it will display *Indicate on screen* in the centre of the block. By clicking on this blue text the Studio will minimize and the top-level window below the Studio will appear. Make sure this is the window you want to let your Robot interact with. Click on the element where you want your robot to type the text into.

Type the text in between quotation marks (“”) in the bar under *Indicate on screen* which says *Text must be quoted*. Now your robot types your text in the chosen field. Example :





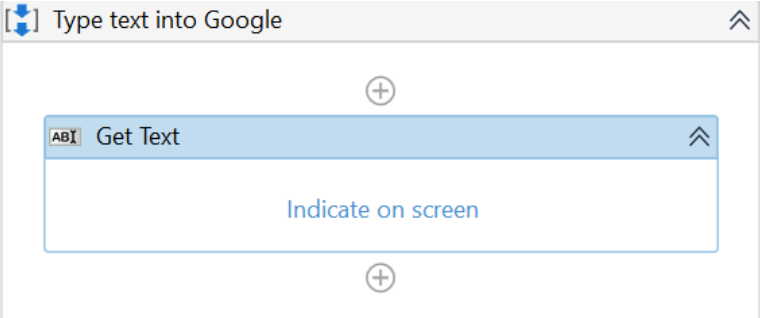


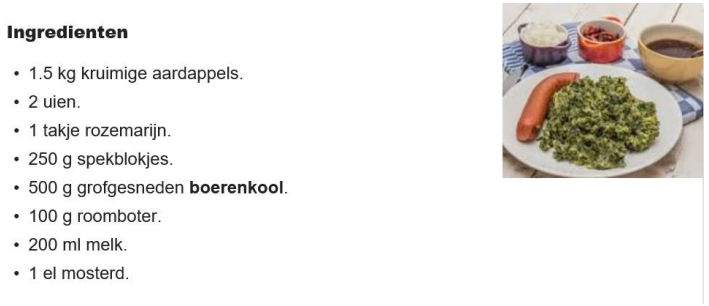
Lesson 5 - Activities

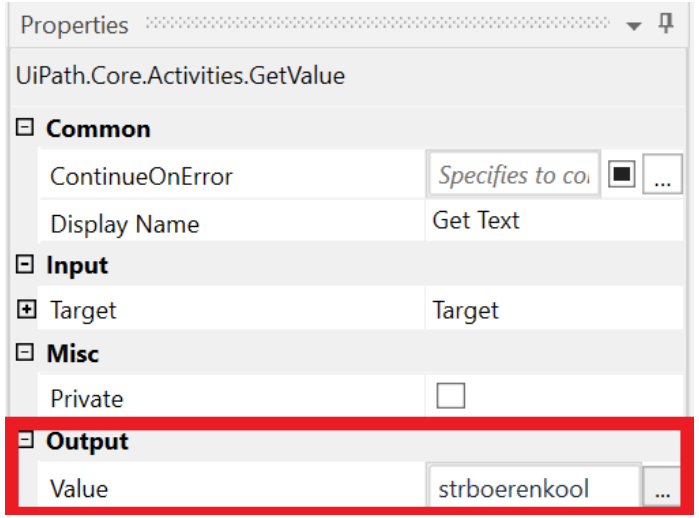
Get Text Activity

Video - Get Text Activity

Retrieves the text from an element within an application or browser and stores it into a string variable. For example you want to get the ingredients for a traditional Dutch recipe: ‘Boerenkool’. After typing this in google you can get the text from this recipe. When dragging this Activity in your Workflow it will display *Indicate on screen* in the centre of the block. By clicking on this blue text the Studio will minimize and the top-level window below the Studio will appear. Make sure this is the window you want to let your Robot interact with. Click on the element where you want your robot to get the text from. In the output property of this activity you can create a string variable that stores this text.







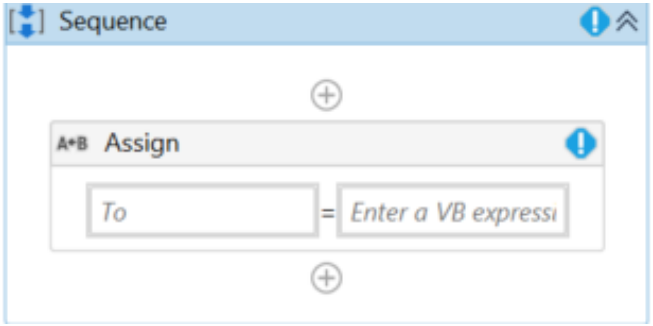
Lesson 5 - Activities

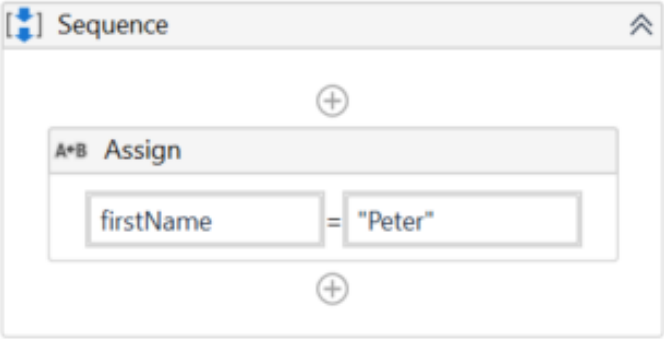
Assign Activity

Video - Assign Activity

The **Assign** activity is an important activity that is going to be used quite often, as it enables you to assign a value to a variable. You can use an **Assign** activity to increment the value of a variable in a loop, sum up the value of two or more variables and assign the result to another variable and a lot more.

In the left box of the assign activity you enter a variable that you're going to assign a value to. In the right box of the activity you enter the value that you want in the variable. An example can be that you want to assign a value "Peter" to the variable firstName.



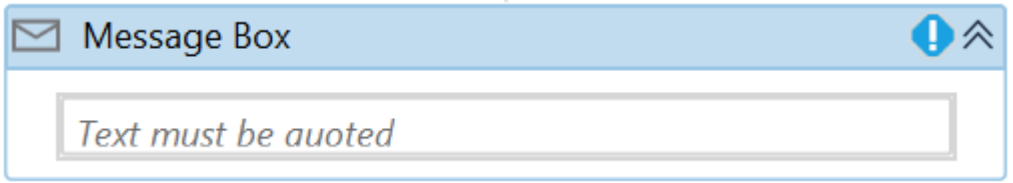


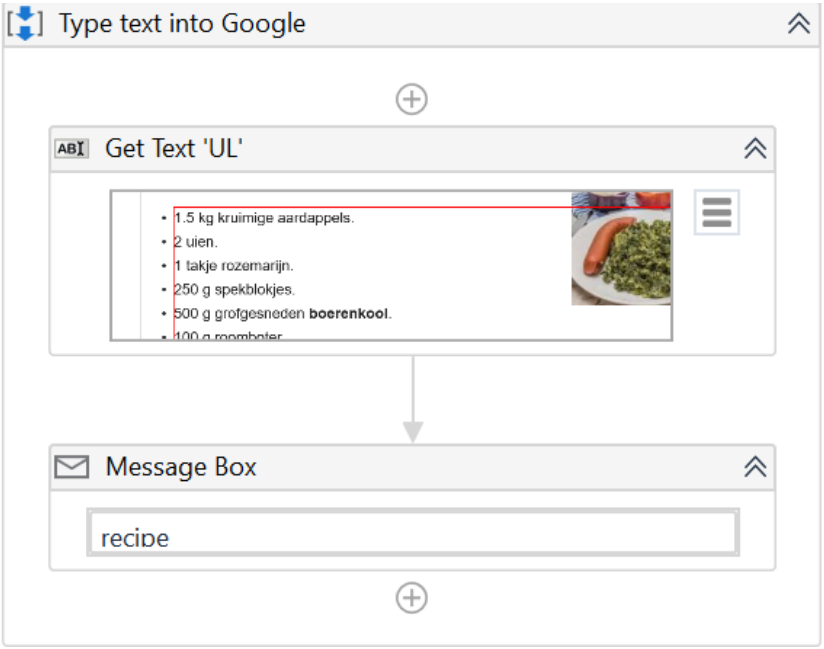
Lesson 5 - Activities

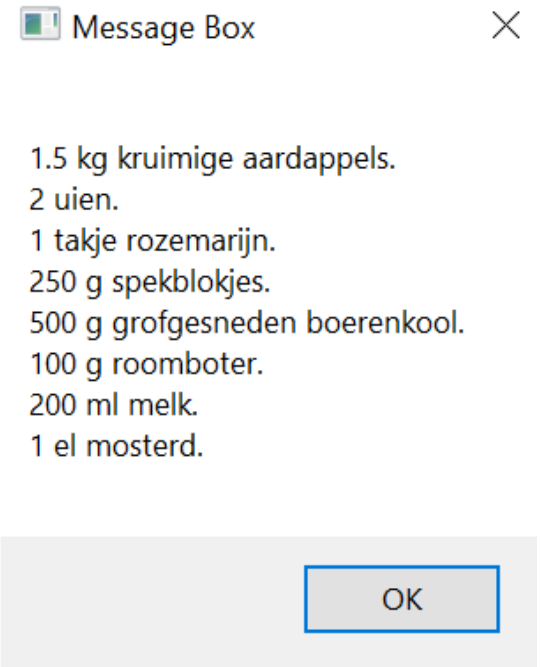
Message Box Activity

Video - Message Box Activity

The **Message Box activity** let you Display a message box with a specific text. This is specifically useful when you want to see the output of an activity or from a variable. For example from the previous get text activity a recipe variable is created that contains the ingredients that you extracted from the 'Boerenkool' recipe. If you want to see if this activity got the right text you can check this with the message box. When dragging this Activity in your Workflow it will display *Text must be Quoted* in the centre of the block. Here you can type the text you want to see or enter the variable you want to display. See the steps below for the 'Boerenkool' example.







Important: A message box can only display String/Text variables!

Video Link

Lesson 5 - Activities

Activities - Exercise

The Get Text Activity is a very useful activity to gather data. In this assignment you are going to retrieve data in the form of text and store it in a variable.

Assignment:

- Build a bot that performs the following actions:

1. Open Google
2. Type “How many legs does a horse have” into the search bar
3. Get the text of the answer and store it in a string variable
4. Use a message box to check if the answer is stored in the variable

Time: 20 minutes

Solution Video

Lesson 5 - Activities

Activities – Exercise 2

What if you don't want to type just one question in the google search bar? Is it possible to use the same workflow for different questions? That's right... You can use a variable for this! Let's see how this works

Assignment

- Create a string variable and give it the name: strQuestionGoogle
- Give it the value "How old is the oldest animal in the world?"
- Insert an open browser, type into and click activity for your workflow, just like you did in the previous exercise
- At the Type into activity you insert the variable strQuestionGoogle
- Run the Robot
- Change the value of your variable to "What is the largest animal on this planet"
- Run the Robot again
- As you can see you can use the same workflow for different values of your variable ☺

Time: 20 minutes

Solution Video

Lesson 5 - Activities

Activities- Recap

Well Done! This chapter contained a lot of information, so let's look back on what we've learned!

You now know how to work with **activities** in UiPath Studio. These are the building blocks of our processes and make our robot execute several actions.

You were introduced to a couple of the mostly used activities which include **Click**, **Open Browser**, **Type Into**, **Get Text**, **Assign** & **Message Box**. If you combine these activities, you are able to build small automation processes which you did in the exercise!

You've built your first robot, congratulations! But what if your robot must make a decision? Can we implement business rules or exceptions within our process? Let's find this out in the next chapter about conditional Activities!

- Activities
- How to work with Activities in UiPath Studio
- Click Activity
- Open Browser Activity
- Type Into Activity
- Get Text Activity
- Assign Activity
- Message Box Activity
- Building your first Robot

Lesson 6: Conditional Activities

Loops & Conditions

Lesson 6 - Conditional Activities

Loops & Conditions – Introduction

This chapter will show you how to work with **Conditional Activities**. In the previous chapter you learned that UiPath has a lot of activities which let your robot perform a lot of actions. But what if your robot needs to take different scenarios into account? Take different actions while facing conditions? For this we use conditional Activities!

This chapter will present you with two videos on conditional activities that introduce you to their usage in UiPath Studio.

At the end of this chapter there will be an exercise again that will challenge you to bring the theory you just learned into practice!

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Lesson 6 - Loops & Conditions

Loops & Conditions – Introducing Conditions

What?

Conditions allow robots to make decisions based on **if-then-else** logic.

Why?

In almost every business process, it is necessary to distinguish various situations, and perform different actions based on that situation.

How?

To implement conditions, you can use the **If** Activity. This Activity contains a condition field and two paths filled with Activities. Based on the condition field, a specific path to be followed is chosen.

Lesson 6 - Loops & Conditions

Conditions – If Statements

Video - Conditional Activities: If

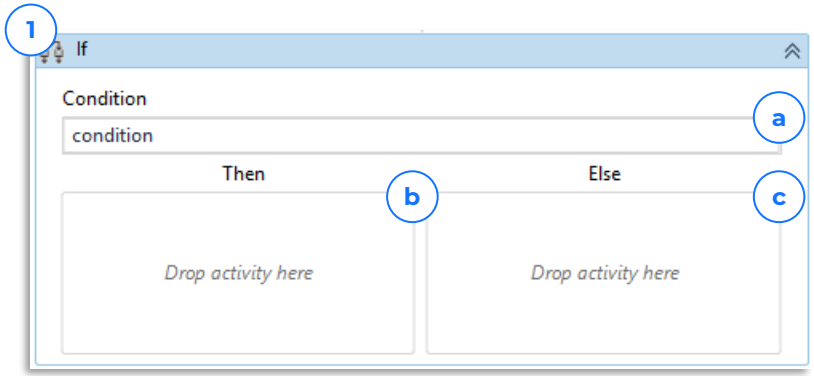
In UiPath, an **If statement** is used to implement Business Logic, also known as Business Rules. The If statement contains two possible paths that can be filled with Activities. Based on a defined condition, the path which is going to be performed is chosen.

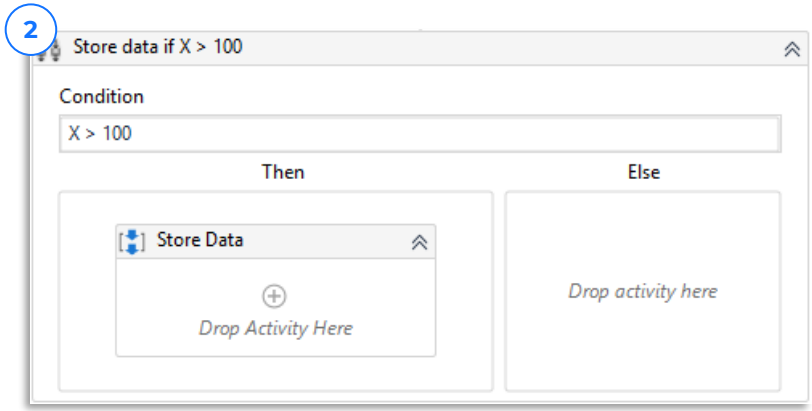
The if statement works as follows:

1. It checks if the *condition* *(a)* is true.
   a. If the condition is true, perform the activities in the *Then*-section *(b)*.
   b. If the condition is false, perform the activities in the *Else*-section *(c)*.

Example Use

- If you want store data when variable 'X' is larger than 100, you include a condition X > 100. Include all the actions to store the data in the Then path and leave the Else path empty. *(2)*
- You use the *Element Exists* Activity to check whether an element is present in an application. You add the If-activity to perform one set of Activities if the element is present, and another set of Activities if the element is not present.

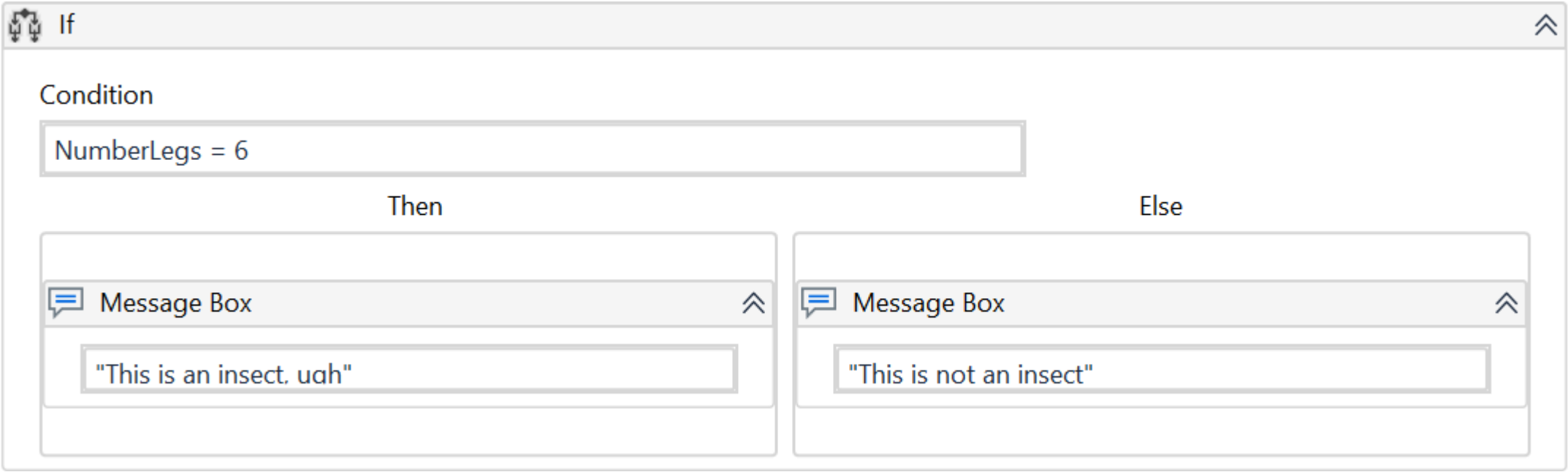




Lesson 6 - Loops & Conditions

Conditions - If-Else Activity Example

You have a list of animal species with the corresponding number of legs of that specie. You want to check whether this animal is an insect or not. This can be checked by checking the number of legs of that animal. Insects always have six legs. So when an animal from your list has 6 legs it can be specified as an insect. If the animal is an insect you want to display “This is an insect, ugh“ and if it is not an insect you want to display “This is not an insect”.



Lesson 6 - Loops & Conditions

Loops – Introducing Loops

What?

Loop activities perform the same set of activities, either a pre-specified number of times (For loops), or until a condition is not met anymore (While loops).

Why?

When a process contains repetitive parts, using Loop activities avoids creating messy workflows.

How?

There are two types of Loops in UiPath: **For** and **While** Loops. The activities that are repeated are dragged into their content blocks.

Lesson 6 - Loops & Conditions

Loops – For Loops

Video - Conditional Activities: Loops

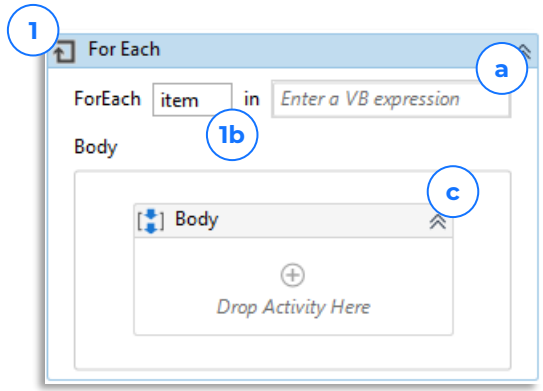
In UiPath, For-loops are executed using the **For Each (1)** or **For Each Row (2)** activities. These activities take a variable containing multiple values as input.

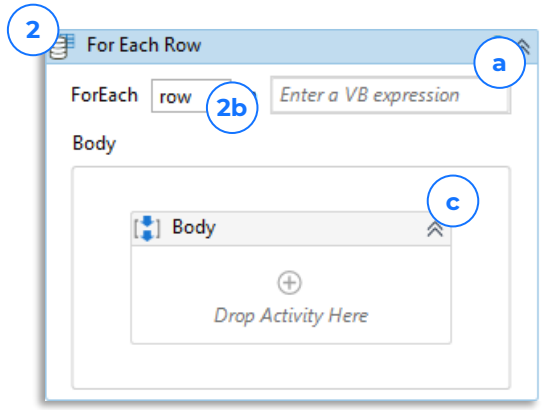
In the **For Each**-activity an array variable can be used, for example. In the **For Each Row** activity, a Data Table variable is used.

For Loops work as follows:

1. Take the *input variable (a)* and split it up into all elements.
2. Pick the next element from the input variable and assign it to the *Item (1b)* or *Row (2b)* variable.
   a. If it is the first time processing, the first element is assigned. If it is the second time, the second element is assigned, etcetera.
3. The activities which have been dragged into the *Body-section (c)* of the For Loop are performed.
4. After the last activity is performed, **jump back to step 2**.

When each value from the input variable has been addressed by the loop Activity it automatically steps out of the block and UiPath will continue with the rest of the Workflow. The For Each Loop is used when you want to perform a certain set of activities a fixed number of times.

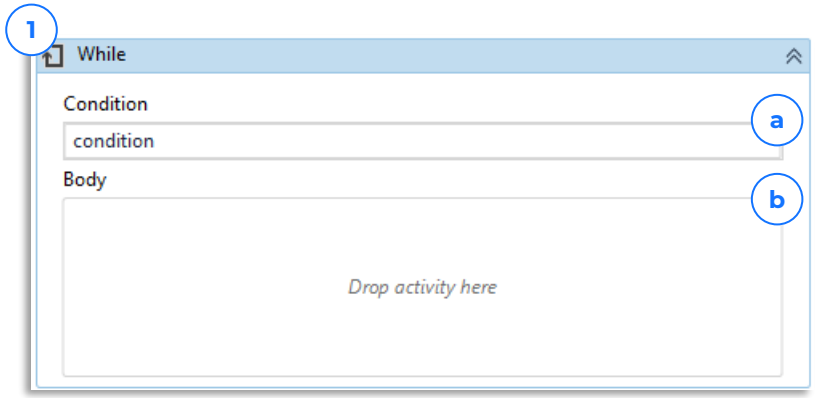


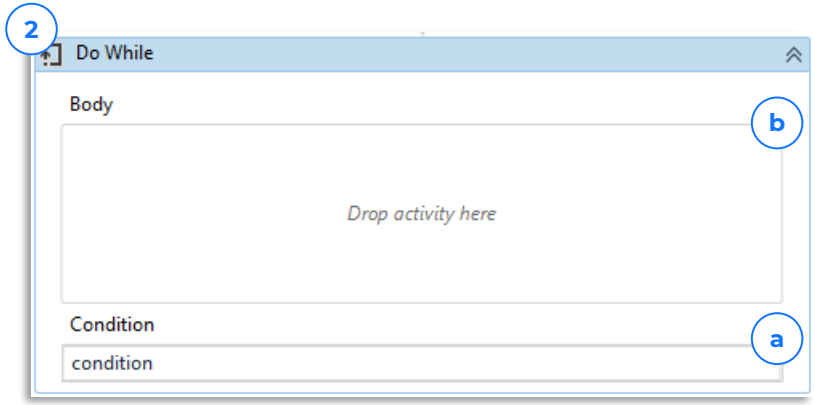


Lesson 6 - Loops & Conditions

Loops – While Loops

In UiPath, While loops can be executed using the **While** Activity or the **Do While** Activity. A While loop performs actions based on a defined condition. The difference between the While and the Do While loop is:





While Loops *(1)* work as follows:

1. Check if the *condition (a)* is true. If it is true, **proceed to step 2**.
2. The activities which have been dragged into the *Body-section (b)* of the While Loop are performed.
3. After the last activity is performed, **jump back to step 1**.

Do-While Loops *(2)* work as follows:

1. The activities which have been dragged into the *Body-section (b)* of the While Loop are performed.
2. Check if the *condition (a)* is true, jump **back to step 1**.

While the condition is met it will iterate through the list of Activities inside the body. When the condition is not met for the first time, it will automatically stop the loop. This implies that the Activities inside the loop must alter at least one bit of information from the condition which could set it to false. Otherwise the loop will continue going forever.

Lesson 6 - Loops & Conditions

Loops - Exercise

Now it's your turn to start working with the For Each and While activities. Try to create a workflow with both the activities as described below.

Assignment

For Each:

1) Create an array containing the names of your top five favorite animals

2) Show the name of each animal in a message box

While:

1) Create the following array {1,2,3,4,5}

2) While the number in the array is <3, show the value in a message box

Time: 20 minutes

Solution Video

Lesson 6 - Loops & Conditions Recap

What did we learn?

Good Job! In this chapter you learned how to handle different business situations within UiPath Studio. You have gotten familiar with making decisions based on **If Statements**, looping through sets of variables with **For Loops** and looping through a set of activities with **While Loops**.

You have learned that an **If Statement** checks if a set condition is met and performs certain activities based on if that condition is met. **For each loops** are ideal for looping through sets of data. Using For loops you can perform a set of activities for each item in the loop. Finally, you have learned how to loop through a set of activities if/until a condition is met, using **While Loops**.

As you know during business processes a lot of times you work with data from excel. But how can you use this data in your workflow and perform certain actions with it? Let's learn more about this in the next chapter about working with Data Tables!

If Statements

For Each Row/Item

(Do-)While Loops

Lesson 7 Data Tables

Introduction to Data Tables

Lesson 7 - Data Tables

Introduction

Hi there! In the previous lesson you've learned how to implement different business exception within your processes. Now A lot of processes contain data from excel. So how can you work with these excel files?

This chapter will learn you the basics of working with excel files and data tables. Watch the introduction video , which will let you become familiar with working with data tables and will give you an idea of what you can use in UiPath Studio.

Introduction video Data Tables

Lesson 7 - Data Tables

Data Tables – Introducing Data Tables

What?

A **data table** is a simple and effective method of **structurally saving data** and is (one of) the most used and easiest method(s) of data structuring to use with RPA. It is made up of columns representing various variables, and rows representing items.

Why?

Practically every process involves data handling or data processing, so even in the simplest processes you will encounter data handling. Therefore, Many processes will have Data Table Transaction data to start with. When working with Excel you are (nearly) always working with data tables and performing actions with data tables.

How?

Using Data Table Activities such as build data table, read range and write range

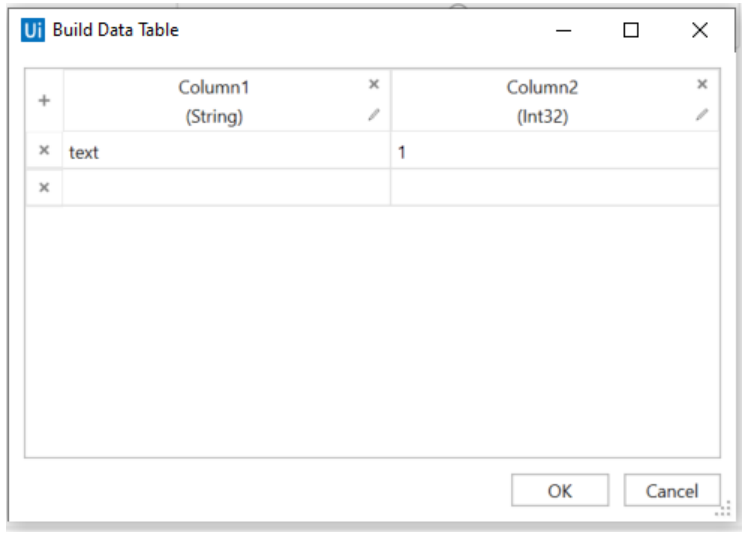
Lesson 7 - Data Tables

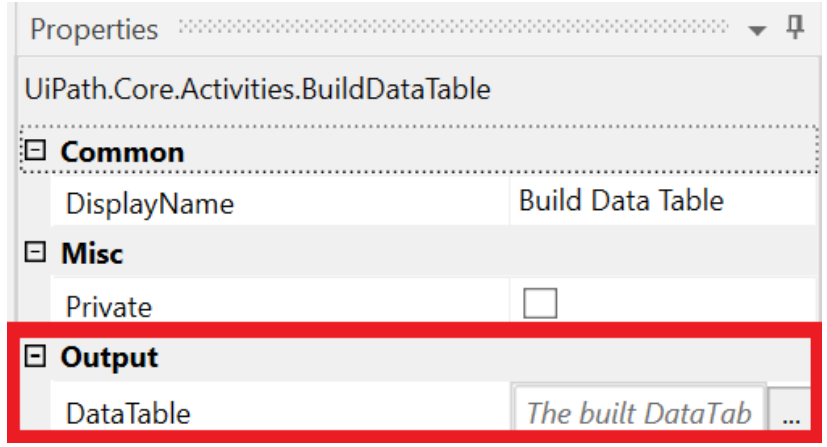
Data Tables - Build Data Table Activity

The **Build Data Table Activity** creates a Data Table according to a specified scheme.

Drag the Build Data Table activity in your workflow such that you can start customizing a Data Table

When you click on the **Data Table box**, the Data Table builder is displayed. Here you can create the Columns you need. With the "**+**" sight on the left you can add a column and give it the right name and datatype. In the rows you can insert the values of that column. In the Properties panel of the BuildDataTable Activity you can create a DataTable variable to store the Data Table in.

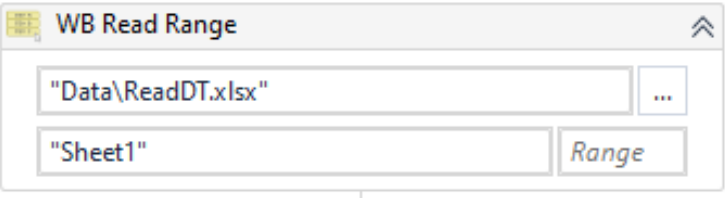




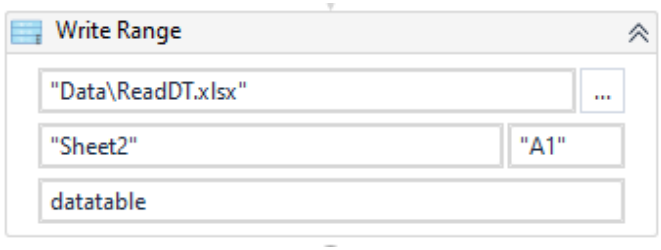
Lesson 7 - Data Tables

Data Tables - Read Range & Write Range Activity

The most used methods of retrieving a data table is either getting it from Excel, or scraping from the web. When you want to let your robot read the data from an Excel you can use the **Read Range** activity. With this activity you read the specified range of an excel file and stores it in a data table variable. When you want to let your Robot create an Excel file you can use **Write Range**. With this activity you create an Excel file from a data table variable.



Place the file path to the Excel workbook and sheet name in the activity. With the range, indicate what part of the sheet you want to include in the outputted data table. If you want the robot to read the whole range of your sheet, leave the range box empty. At the property section you can create a DataTable variable in the output panel so that the data is stored inside a data table.



Place the file path to the Excel work book and sheet name in the activity. With the starting cell, indicate where the data table should start. In case your data table includes headers, the headers will start in that cell.

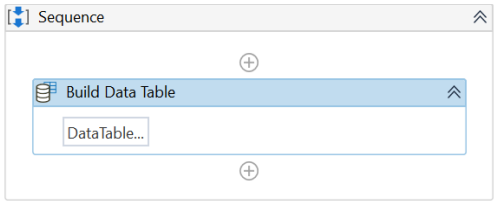
Lesson 7 - Data Tables

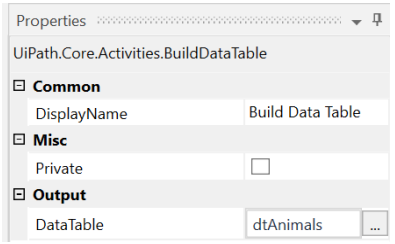
Data Tables – Visualize your Data Table

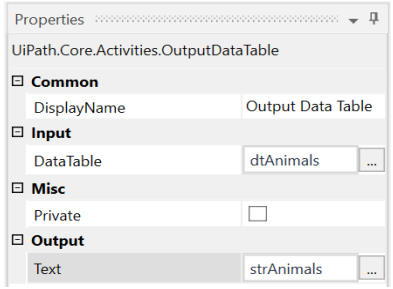
If you want to see the output of your Data Table you can use the **Output Data Table activity**. This activity converts a Data Table variable to a string variable

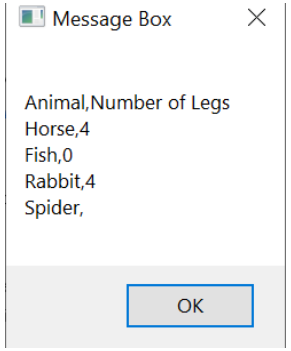
You can print this string variable such that you can see the content of the Data Table.

Use the Message Box activity to display the output of your Data Table









Lesson 7 - Data Tables

Exercise

The best way to get familiar with building and reading DataTables is by doing it yourself. In this assignment you're going to get familiar with DataTables. Good Luck!

Assignment

- Build a Datatable containing 2 Columns and 3 Rows:

1. Species
2. Number of Legs

- Use the write Range Activity to write the whole datatable to an Excel file
- Use the Read Range Activity to store the whole Excel file in a new datatable

- Use the output Datatable Activity and a Message Box to show the value of the Data Table

Time: 20 minutes

Solution Video

Lesson 7 - Data Tables - Recap

What did we learn?

Good Job! In this chapter you learned the basics of DataTables and how to use them. You have gotten familiar with building DataTables, reading them and how you can extract and display the data stored inside.

The Build Data Table activity allows you to create a data table by your own. You can create as much columns and rows as you want and store every type of data within. If you want to write a data table to an Excel sheet, you use the **Write Range** activity. When you want to do the opposite, read a data table from Excel, you must use the **Read Range** activity. And finally, when you want to extract the data from the data table, and use it for example in a message box, the **Output Data Table** activity is used.

As you may know, UiPath works on interaction with small elements of code. Each application or interface UiPath works with, is built out of those elements. But how does UiPath interact with those little elements? That is done with the use of **Selectors**. Next lesson you will learn everything about selectors and how to work with them.

Build Data Table

Read Range

Write Range

Output Data Table

Lesson 8 Selectors

Creating Unique Selectors

Lesson 8 - Selectors

Introduction

Good Job! You now know how to work with Excel files and how to use data tables within UiPath Studio.

In this chapter we're going to introduce you to a more advanced development concept, namely selectors. When you let your robot work with different UI elements it needs the location of these elements to know where to find it.

How does this work? Watch the introduction video that will explain what a selector is.

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Introduction video Selectors

Lesson 8 - Selectors

Selectors– Introducing Selectors

What? A selector is the code description of a UI Element, which is used by the robot as a unique reference to a specific Element in an application

Why? A selector allows a robot to interact with an application, regardless of changes in looks or lay-out

How? With the UI Explorer in UiPath Studio

Lesson 8 - Selectors

Selectors - What is a Selector?

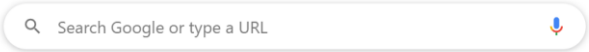
- To automate specific actions in the user interface, you are required to interact with various windows, buttons, drop-down lists and many others

- A **Selector** is a (code) description of UI Elements. The selector is used by your Robot as a handlebar to find a specified UI element inside windows.
- These windows contain loads of UI Elements. Examples of these elements are a search button, a login fields and text.

- It is important that this description is unique, because otherwise the robot does not understand what part of the window you need for the action.

- All the information you see when you open the Selector Editor is directly derived from the source code of the webpage or application. Ideally you let UiPath automatically create the selector by indicating the element on screen by clicking. Sometimes it is possible that there are identical elements on screen, but you only want to let your Robot interact with one of them specifically. An automatically generated selector might not be the solution then. In this case you need to rebuild the selector in such a way that it is unique and points to the right UI element that you need.

```
<html app='chrome.exe' title='Facebook - Meld je aan of registreer je' />
<webctrl id='email' tag='INPUT' />
```



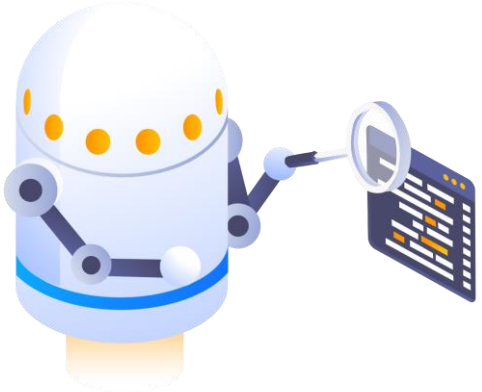

```
<html app='chrome.exe' title='RPA Academy' />
<webctrl tag='BUTTON' class='btn btn-outline-danger btn-sm' />
```

Lesson 8 - Selectors

Selectors - Full and Partial Selectors

There is a difference between **full selectors** and **partial selectors**. **Full selectors** include information about the application that is being used and the specific window within the application that is opened and adding to that the information that leads to the desired UI element.

Partial selectors only lead to the desired UI element without any information about the application used. Full selectors are in principle more stable. Only use partial selectors when you want to build modular Workflows that you can use on different windows containing similar UI elements. (We will discuss this later in the course)



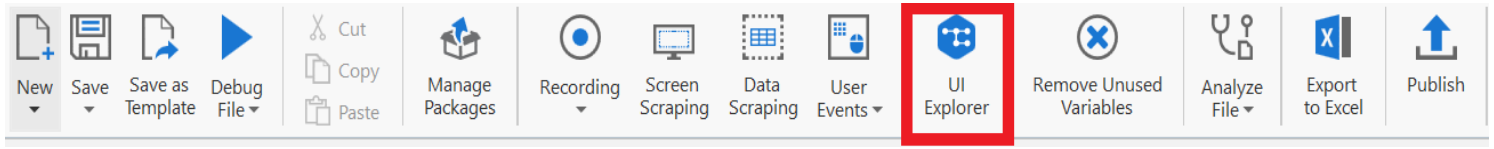
Lesson 8 - Selectors

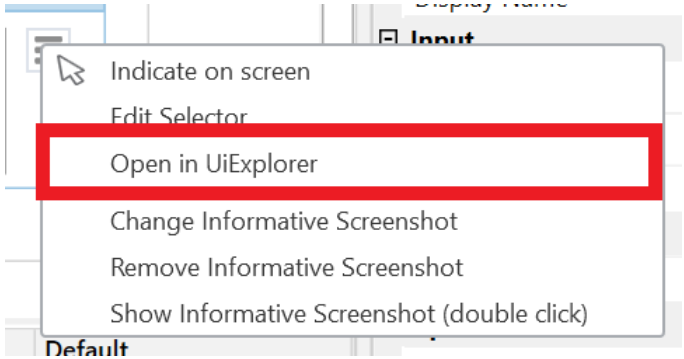
Selectors - Building a reliable Selector

It is important to create a unique selector for UIElements in such a way that your robot can recognize this selected field in a window. Sometimes just adding the index in which the element is ranked within the window is enough to distinguish between elements.

When you need to reshape your selector such that it is unique you can use the UI Explorer.

The **UI Explorer** can be used to inspect the UI element and tweak the selector accordingly. You can find the UI Explorer in the toolbar or in the menu field of your activity





Important here is the validate button. Every time you changed something in the selector, you should validate if this selector is working for your specific Ui Element. Use the highlight button to check whether the right element is selected.

Lesson 8 - Selectors

Selectors - Important guidelines when building selectors

There is no one way to fix all selector related issues as different situations require different solutions. Nonetheless, there are a few guideless which you can incorporate in your selector fixing strategy:

- Avoid using **CSS-selectors**. These selectors tell the Robot the hierarchy at which the UI element is located within the entire application. You will see something like *Body > Div > Div > ... > Div >* etc. A *Div* is a page division. Since most applications are often submissive to changes, one change anywhere within the window will result in an invalid selector, even if the changes had nothing to do with the UI element you selected.
- Try to incorporate the **ID selector** as the ID of an UI element is most of the time unique. When you see for example that the ID selector has a value of '1' you can imagine that this is not really a unique value. When this is the case or when there is no ID selector available the **Class selector** is a good second option. Important with the Class selector is that it can never go without any other type of selectors as classes, as the name already gives away, are assigned to group of elements.

Lesson 8 – Selectors - Recap

What did we learn?

Good Job! You have finished the lesson on Selectors within UiPath. In this lesson you have learned the basics about Selectors, what they are and how the are selected. Later in the Ciphix Academy, you will learn more about selectors. But for now you should be all set to go and create your first selectors.

In this lesson, you have learned what **selectors** are and where they are used for. You have learned the differences between **Full** – and **Partial Selectors** and what the benefits of each type are. A selector can be specified using a tool integrated in UiPath, this is the **UI Explorer t**ool. Inside this tool you can dive into a selector and all its underlying parts.

Now you are familiar with all basic activities and all additional aspects, you have should be able to create your first real process. But how do you build a process? In next lesson, you will learn how you can create your first sequence and flowcharts.

Selectors

Full – vs. Partial Selectors

UI Explorer

Lesson 9: Sequences & Flowcharts

Introduction to Workflow structures

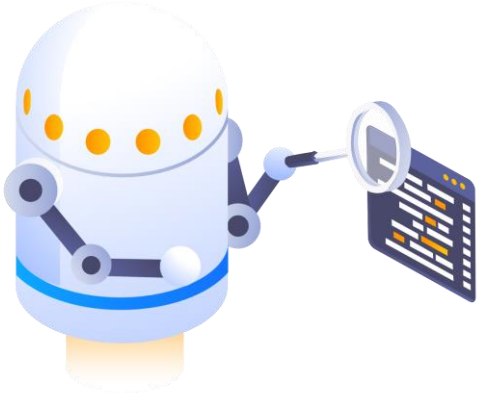
Lesson - Sequences & Flowcharts

Introduction

You're almost there! Now you have completed the lessons on all basics regarding creating your own process, you're going to learn how you use this in sequences and flowcharts.

In this chapter we're going to introduce you to working with sequences and flowcharts. These are used to store activities that are performed by the robot. An automation project consists of a combination of multiple sequences and flowcharts.

How do sequences and flowcharts work?



Lesson 8 – Sequences & Flowcharts

Sequences & Flowcharts– Introducing Sequences and Flowcharts

What?

Within workflow development, two structures can be distinguished: Sequences and Workflows. Both have activities stored inside them that are used during a process.

Why?

It is important to create a structured and a clear workflow. Sequences and Flowcharts help you structuring your workflow and activities.

How?

Sequences and Workflows can be seen as containers that have multiple activities within them. The containers give a structured overview of the activities inside them and decide the way these activities are performed.

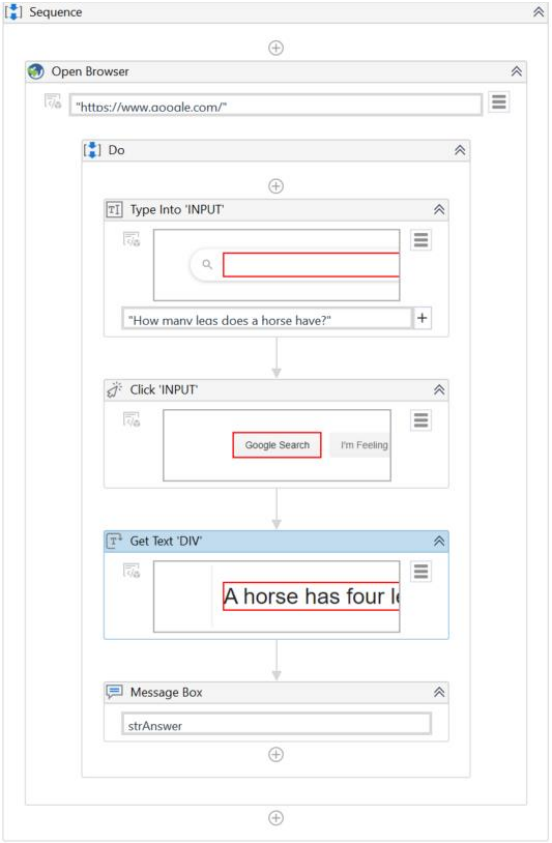
Lesson 9 - Sequences & Flowcharts

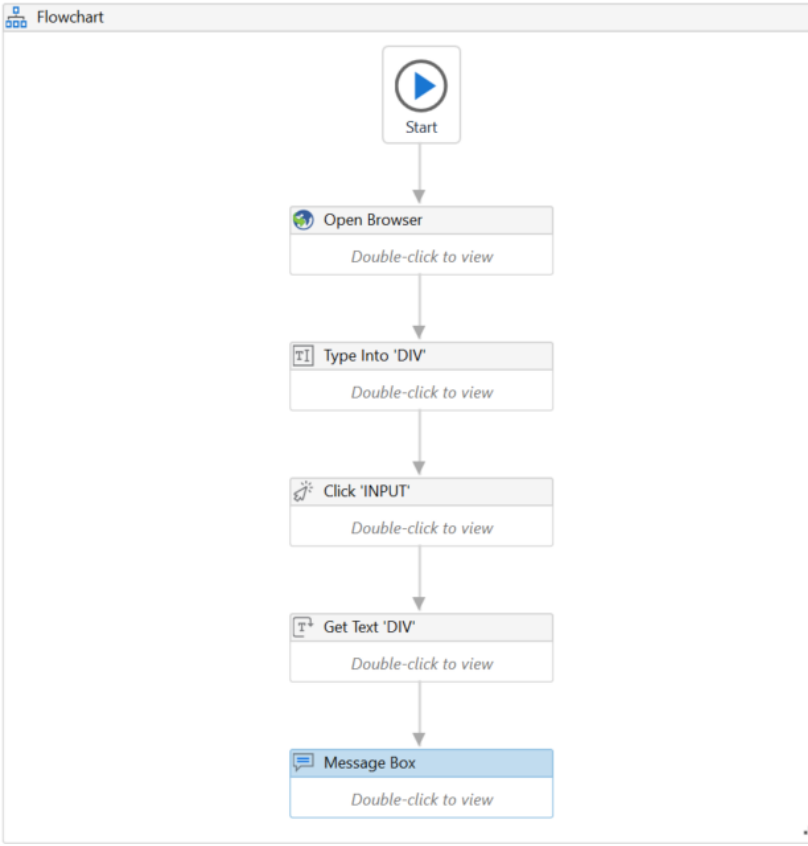
Sequences & Flowcharts - Definitions

- When designing workflows, the developer must make a choice of how the canvas of the workflow is structured.

- Within workflow development, two structures can be distinguished: **Sequences and Flowcharts**. You can see these activities as containers that have multiple other activities inside them.

- Sequences and Flowcharts are used for the same goal but in different situations.

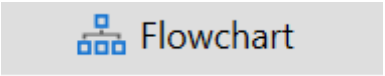




Lesson 9 - Sequences & Flowcharts

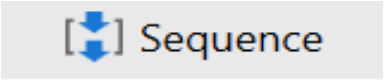
Sequences & Flowcharts - Difference

Flowcharts



- Enables to adjust easily no matter how complex the workflow gets, making it more flexible
- Enables developers to keep the workflow much cleaner and organized
- **However**, unnecessary when activities follow each other in linear fashion/When there are no conditional decisions made.

Sequences



- Preferred when activities follow each other in linear fashion
- **However,** unsuitable for workflows that contain many nested conditional statements (When using a lot of If in if activities)

	Flowchart	**Sequence**
Number of nested conditional statements (nested in a single conditional statement)	More than two	Two or less
Structure of activities	Non-linear (multiple branches)	Linear

Lesson 9 – Sequences & Flowcharts - Recap

What did we learn?

Congratulations, you have finished the last lesson in the Ciphix RPA Academy – RPA Applicants Course. In this introduction you were introduced to structuring your workflows using Sequences and Flowcharts.

You have learned the differences between these two activities and when you should use them. Each of this activities has its benefits. Sequences are best used in structured, minimal conditional and linear workflows. Flowcharts are best used in workflows with multiple nested conditional statements and the more complex sequences of activities.

Selectors

Full – vs. Partial Selectors

UI Explorer

Congratulations!

You have successfully finished the RPA Applicants Course of the Ciphix RPA Academy. You have learned all the basics of RPA and you are at the beginning of becoming an RPA expert!

You are now ready to make the RPA Case. In this assignment you can apply the knowledge that you have gathered in this course. Don't forget that you can always find additional explanations on the internet by searching on the UiPath documentation website and by means of Google!

Good luck and we are looking forward to your RPA Solution!

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